

Fold & Fire

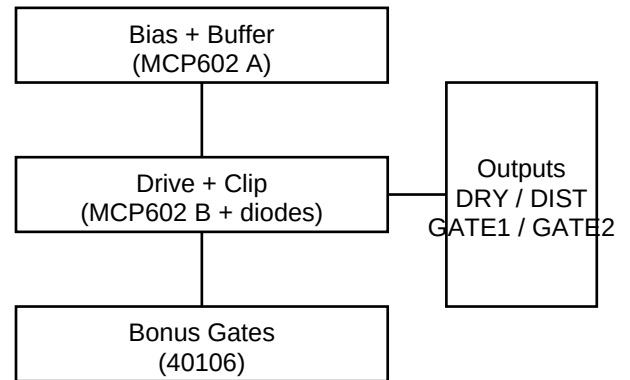
AE Modular BRAEDBOARD • MCP602 diode distortion + bonus gates (40106)

Vref (~2.5V) lets audio live on a 5V supply. Buffer the biased signal, then drive it into diode feedback so peaks compress and harmonics appear. Bonus: a 40106 turns DIST into gates (edges/threshold decisions).

Build in layers

- 1) Vref: 100k/100k + 100nF.
- 2) Bias input: 100nF coupling + 100k to Vref.
- 3) Buffer (MCP602 A) → DRY.
- 4) Drive (MCP602 B) gain around Vref.
- 5) Clip: anti-parallel diodes in feedback.
- 6) Bonus: 40106 → GATE1/GATE2 from DIST.

Concept map



Checkpoints

- Vref ~2.5V stable.
- DRY idles near Vref.
- DIST gets louder with gain.
- Diodes add harmonics/compression.
- GATE1 toggles cleanly (LED).
- Keep gate wires away from audio.

Patch recipes

- Parallel dirt: blend DRY + DIST.
- Timbre→Rhythm: GATE1 triggers envelopes.
- Gate2 (filtered) gives slower pulses.
- Asymmetry = character.

Build hygiene

Decouple each IC (100nF close). Keep Vref clean. Keep the clipper (–) node short. Route gate wiring away from audio nodes to avoid ticking.